

Math 63: Real Analysis

Prishita Dharampal

Problem 1. Let $a_1, a_2, a_3, \dots$ be a sequence of real numbers. Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then so does $\sum_{n=1}^{\infty} (a_n)^p$ for any integer $p > 1$.

Solution.

Since $\sum_{n=1}^{\infty} |a_n|$ converges absolutely, $\lim_{n \rightarrow \infty} |a_n| = 0$ (Corollary 1m page 142). That is there exists a positive integer N , such that for all $n > N$

$$(|a_n|)^p \leq |a_n| < 1$$

since $p > 1$ and $|a_n| < 1$. Then, since $\sum_{n>N}^{\infty} |a_n|$ converges, so does $\sum_{n>N}^{\infty} (|a_n|)^p$ (Comparison Test). Let $\sum_{n>N}^{\infty} (|a_n|)^p = A$. Then we have,

$$\sum_{n=1}^{\infty} (|a_n|)^p = \sum_{n=1}^N (|a_n|)^p + \sum_{n>N}^{\infty} (|a_n|)^p = \sum_{n=1}^N (|a_n|)^p + A$$

Since both $\sum_{n=1}^N (|a_n|)^p$ and A are finite, the series converges. Note that $(|a_n|)^p = |(a_n)^p|$. Hence $\sum_{n=1}^{\infty} |(a_n)^p|$ is convergent, and $\sum_{n=1}^{\infty} (a_n)^p$ converges absolutely.

Problem 2. Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function, and let $y_1, y_2, y_3, \dots$ be a sequence of real numbers that converges to 0. Prove that

$$\lim_{n \rightarrow \infty} \int_0^1 F(x, y_n) \, dx = \int_0^1 F(x, 0) \, dx.$$

Solution.

Since convergent sequences of points are bounded (page 47), let $M = \text{l.u.b. } |y_n|$. We now look at the restriction of F to the compact subspace $[0, 1] \times [-M, M]$. F is continuous on this compact space, and hence is uniformly continuous. That is for any $\varepsilon > 0$ there exists some $\delta > 0$ such that if the Euclidean distance $\|(x, y_n) - (x, 0)\| < \delta$ then

$$|F(x, y_n) - F(x, 0)| < \varepsilon$$

But note that $\|(x, y_n) - (x, 0)\| = \sqrt{(x - x)^2 + (y_n - 0)^2} = |y_n|$. Then as $|y_n| \rightarrow 0$, given any $\varepsilon > 0$, for large enough N , $|y_n| < \delta$ for all $n > N$ and $|F(x, y_n) - F(x, 0)| < \varepsilon$. Since δ depends only on ε and not on x the bound $|F(x, y_n) - F(x, 0)| < \varepsilon$ holds for all $x \in [0, 1]$. Hence, $F(x, y_n)$ converges uniformly to $F(x, 0)$ on $[0, 1]$. Then by the Theorem on page 138, we have

$$\lim_{n \rightarrow \infty} \int_0^1 F(x, y_n) \, dx = \int_0^1 \left(\lim_{n \rightarrow \infty} F(x, y_n) \right) \, dx = \int_0^1 F(x, 0) \, dx$$

Problem 3. A curve f in $\mathbb{R}^N$ is a continuous function

$$f : [0, 1] \rightarrow \mathbb{R}^N, \quad f(t) = (f_1(t), \dots, f_N(t)) \in \mathbb{R}^N.$$

Let $\mathcal{C}$ denote the set of curves in $\mathbb{R}^N$ with the property that each component function f_j is differentiable on $(0, 1)$ with derivatives f'_j that extend to continuous functions on $[0, 1]$.

For two curves $f, g \in \mathcal{C}$ define the real number $d(f, g)$ by

$$d(f, g) = \sum_{j=1}^N D(f_j, g_j) + D(f'_j, g'_j).$$

Here $D(\cdot, \cdot)$ is as in Chapter IV.6. Then $d : \mathcal{C} \times \mathcal{C} \rightarrow \mathbb{R}$ satisfies the axioms of a metric (you do not have to prove this).

- (a) Prove that $\mathcal{C}$ with the metric above is a complete metric space.
- (b) The length of a curve is defined as in Problem VI.20. Denote the length of a curve $f \in \mathcal{C}$ by $L(f)$. Prove that the function

$$L : \mathcal{C} \rightarrow \mathbb{R}$$

is continuous.

Solution.

Note that $d(f, g) = \sum_{j=1}^N (\max\{|f_j(p) - g_j(p)| : p \in [0, 1]\} + \max\{|f'_j(p) - g'_j(p)| : p \in [0, 1]\})$

- (a) We want to show that any Cauchy sequence of functions (f_n) in $\mathcal{C}$ converges.

Let (f_n) be an arbitrary Cauchy sequence of functions in $\mathcal{C}$. Then we know that for all $\varepsilon > 0$ there exists an M such that $d(f_n, f_m) < \varepsilon/2N$ for all $n, m > M$. That is,

$$d(f_n, f_m) = \sum_{j=1}^N (D(f_{n_j}, f_{m_j}) + D(f'_{n_j}, f'_{m_j})) < \frac{\varepsilon}{2N}$$

That is $\forall j, \forall n, m > M$, $D(f_{n_j}, f_{m_j}) < \frac{\varepsilon}{2N}$ and $D(f'_{n_j}, f'_{m_j}) < \frac{\varepsilon}{2N}$, hence each individual sequence (f_{n_j}) and (f'_{n_j}) is Cauchy. Note that these functions are defined from $[0, 1]$ to $\mathbb{R}$. Since $\mathbb{R}$ is complete, we know that the sequences converge uniformly (Proposition, page 86). Let the convergent functions and derivatives be denoted by f_j, h_j .

$$(f_{n_j}) \rightarrow f_j, \quad (f'_{n_j}) \rightarrow h_j$$

Now we want to show that $f'_j = h_j$, for a given j . Note that each f_{n_j} has a continuous derivative (since it is in $\mathcal{C}$), and we showed above that the sequence (f_{n_j}) and the sequence

of their continuous derivatives converge uniformly. Then the derivative $\left(\lim_{n \rightarrow \infty} f_{n_j}\right)' = f'_j$ exists and is equal to $\lim_{n \rightarrow \infty} f'_{n_j} = h_j$ (Theorem, Page 140). Now we can define the limit f of our original sequence (f_n) ,

$$f : [0, 1] \rightarrow \mathbb{R}^N, \quad f(t) = (f_1(t), \dots, f_N(t))$$

Note that $f \in \mathcal{C}$ because each component f_j is differentiable with derivatives f'_j that are continuous on $[0, 1]$ since it is the uniform limit of continuous functions.

Finally, we verify that $d(f_n, f) \rightarrow 0$. Since f_{n_j} converges uniformly to f_j and f'_{n_j} converges uniformly to f'_j for each j , given any $\epsilon > 0$ there exists M such that for all $n > M$ and all j ,

$$D(f_{n_j}, f_j) < \frac{\epsilon}{2N} \quad \text{and} \quad D(f'_{n_j}, f'_j) < \frac{\epsilon}{2N}.$$

Therefore,

$$d(f_n, f) = \sum_{j=1}^N (D(f_{n_j}, f_j) + D(f'_{n_j}, f'_j)) < \sum_{j=1}^N \frac{\epsilon}{2N} + \frac{\epsilon}{2N} = \epsilon.$$

Hence (f_n) has a limit in $\mathcal{C}$. Since (f_n) was arbitrary, every Cauchy sequence has a limit in $\mathcal{C}$, and hence $\mathcal{C}$ is complete.

(b) The length of a curve f (Problem VI.20) is

$$L(f) = \int_0^1 \sqrt{(f'_1(x))^2 + \dots + (f'_N(x))^2} dx = \int_0^1 \sqrt{\sum_{j=1}^N (f'_j)^2} dx$$

We want to show that L is continuous, i.e., for any $\epsilon > 0$, there exists a $\delta > 0$ such that if $d(f, g) < \delta$ then $|L(f) - L(g)| < \epsilon$.

$$\begin{aligned} |L(f) - L(g)| &= \left| \int_0^1 \sqrt{\sum_{j=1}^N (f'_j(x))^2} dx - \int_0^1 \sqrt{\sum_{j=1}^N (g'_j(x))^2} dx \right| \\ &= \left| \int_0^1 \left(\sqrt{\sum_{j=1}^N (f'_j(x))^2} - \sqrt{\sum_{j=1}^N (g'_j(x))^2} \right) dx \right| \\ &\leq \int_0^1 \left| \sqrt{\sum_{j=1}^N (f'_j(x))^2} - \sqrt{\sum_{j=1}^N (g'_j(x))^2} \right| dx \end{aligned}$$

Applying the Proposition on Page 37 to the Euclidean norm gives us

$$\int_0^1 \left| \sqrt{\sum_{j=1}^N (f'_j(x))^2} - \sqrt{\sum_{j=1}^N (g'_j(x))^2} \right| dx \leq \int_0^1 \sqrt{\sum_{j=1}^N (f'_j(x) - g'_j(x))^2} dx$$

Note that $\sum_{j=1}^N a_j^2 \leq \sum_{j=1}^N |a_j|^2 \implies \sqrt{\sum_{j=1}^N a_j^2} \leq \sum_{j=1}^N |a_j|$. Then, we have

$$\begin{aligned} \int_0^1 \left| \sqrt{\sum_{j=1}^N (f'_j(x))^2} - \sqrt{\sum_{j=1}^N (g'_j(x))^2} \right| dx &\leq \int_0^1 \sum_{j=1}^N |f'_j(x) - g'_j(x)| dx \\ &\leq \int_0^1 \sum_{j=1}^N D(f'_j, g'_j) dx \\ &\leq \sum_{j=1}^N D(f'_j, g'_j) \leq d(f, g) < \varepsilon \end{aligned}$$

Thus $|L(f) - L(g)| < d(f, g)$, so taking $\delta = \varepsilon$, we have that if $d(f, g) < \delta$, then $|L(f) - L(g)| < \varepsilon$. Hence $L : \mathcal{C} \rightarrow \mathbb{R}$ is a continuous function.